

Introduction to PDEs

1 Air Quality Modeling

- problem and model
- the transport equation
- operator notation

2 Diffusion

- modeling the change in concentration
- the Laplacian
- the advection diffusion equation

MCS 472 Lecture 36
Industrial Math & Computation
Jan Verschelde, 13 April 2026

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air quality modeling

Consider the following context.

- An industrial plant emits noxious fumes.
- Suppose once an hour, the plant emits fumes for a few minutes.
- One mile away is our house.

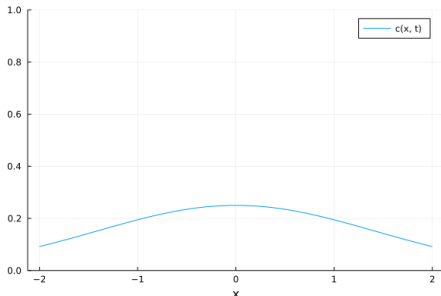
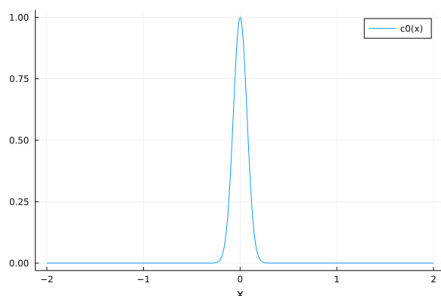
Problem:

*If the wind is blowing from the plant to the house,
what will the maximum concentration of the noxious fumes be
when they reach your house?*

From chapter 2 of *Industrial Mathematics: A Course in Solving Real-World Problems* by Avner Friedman and Walter Littman, published by SIAM in 1994.

evolution of the fume profile

Assuming bell shaped profiles of the fumes:



- $c_0(x)$ is the profile of the fumes produced at time zero, shown in the picture above at the left.
- $c(x, t)$ is the profile of the fumes at place x at time t , shown in the picture above at the right.

a proposed solution

Assume the wind blows at constant speed v towards our house.

The speed multiplied by time traveled equals distance traveled.

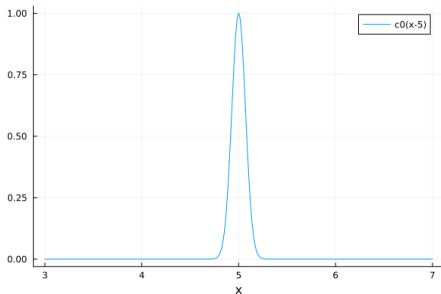
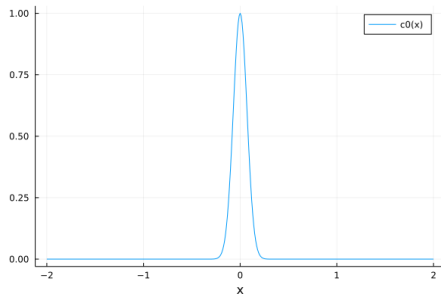
So $v t$ is the distance traveled at speed v at time t .

The proposed solution is then

$$c(x, t) = c_0(x - v t).$$

The original profile of the fumes shifts in the *opposite* direction of the wind, whence the minus sign in $x - v t$.

the shifted fume profile



- At the left is the plot of $\exp(-100x^2)$.
- At the right is the plot of $\exp(-100(x-5)^2)$.

the advection equation

The derivatives of the proposed solution

$$c(x, t) = c_0(x - vt)$$

with respect to space and time are

$$\frac{\partial}{\partial x} c(x, t) = c'_0(x - vt) \cdot 1,$$

$$\frac{\partial}{\partial t} c(x, t) = c'_0(x - vt) \cdot (-v).$$

What follows behind the $\cdot$ after c'_0 comes from the chain rule.

Eliminating $c'_0(x - vt)$ yields *the advection equation*, which is:

$$\frac{\partial c}{\partial t}(x, t) + v \frac{\partial c}{\partial x}(x, t) = 0, \quad c(x, 0) = c_0(x).$$

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the transport equation

The Partial Differential Equation (PDE) we just derived

$$\frac{\partial c}{\partial t}(x, t) + v \frac{\partial c}{\partial x}(x, t) = 0, \quad c(x, 0) = c_0(x),$$

is also known as *the transport equation*.

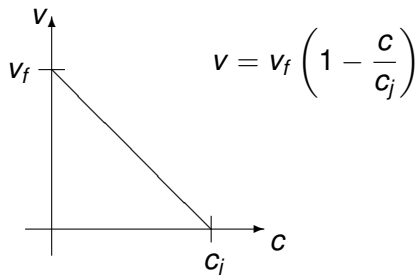
$c(x, t)$ is the concentration at location x at time t ,
traveling at constant speed v .

Application: traffic where $c(x, t) = \frac{\text{\# vehicles}}{\text{unit mile}}$.

speed in function of the concentration

$c(x, t) = \frac{\text{\# vehicles}}{\text{unit mile}}$, traveling at speed v .

Traffic slows down when the number of vehicles increases.



To model traffic flow, v depends on the concentration.

- v_f is the *free traffic speed*, when the road is free.
- c_j is the *jam density*, $v = 0$ at a traffic jam.

the transport equation with variable speed

The variable in the transport equation is the concentration $c(x, t)$ and the partial differential equation has two terms:

- 1 the change in concentration c in time t ,
- 2 the change in concentration c in space x .

With variable speed v , the equation is then

$$\frac{\partial c}{\partial t}(x, t) + \frac{\partial(v c)}{\partial x}(x, t) = 0, \quad c(x, 0) = c_0(x).$$

Pulling v into the $\frac{\partial}{\partial x}$ term agrees with $c(x, t) = c_0(x - v t)$, proposed in the traveling profile of the fumes in the air quality model.

We study traffic flow in greater detail in a later lecture.

an exercise

Exercise 1:

Let $f(x)$ be continuously differentiable and consider

$$\frac{dx}{dt} = f(x), \quad t > 0, \quad x(0) = x_0.$$

Its solution $x(t)$ depends on x_0 , so we write $x(t, x_0)$ as the unique curve that passes through the point $(x_0, 0)$.

Show that $x(t, x_0)$ is differentiable and that

$$z(t) = \frac{\partial x(t, x_0)}{\partial x_0}$$

satisfies

$$\frac{dz}{dt} = f_x(x(t, x_0))z, \quad f_x = \frac{df}{dx}, \quad z(0) = 1.$$

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operator notation

In moving to three dimensions, operators notations are useful.

$c(x_1, x_2, x_3, t)$ is the concentration at the point (x_1, x_2, x_3) , at time t .

The velocity vector of the transport is

$$\vec{v}(x_1, x_2, x_3, t) = v_1 \mathbf{i} + v_2 \mathbf{j} + v_3 \mathbf{k},$$

where v_1, v_2, v_3 are functions of (x_1, x_2, x_3, t) .

Definition (the del operator)

The *del operator* is $\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$.

the three dimensional advection equation

The three dimensional advection equation is

$$\frac{\partial c}{\partial t} + \nabla \cdot (\vec{v} c) = 0,$$

formulated with the del operator

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

and the dot product

$$\nabla \cdot \vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}.$$

a cosine hill

Let (x_0, y_0) be the origin of the fumes.

Consider the radius R , with

$$R^2 = (x - x_0)^2 + (y - y_0)^2.$$

Then the initial concentration

$$c_0(x, y) = \begin{cases} 50 \left(1 + \cos \left(\frac{\pi R}{4} \right) \right) & \text{if } R < 4, \\ 0 & \text{otherwise,} \end{cases}$$

is referred to as a cosine hill.

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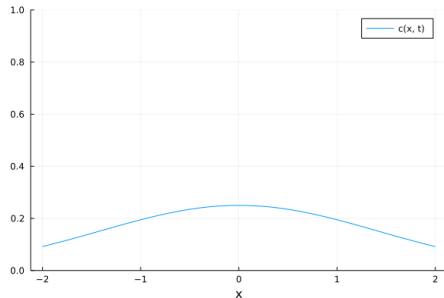
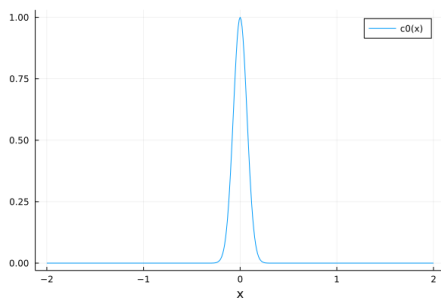
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modeling the change in concentration

Consider the evolution of the profile of fumes:



The transition from high to low concentration is *diffusion*.

a symbolic calculation

```
>>> from sympy import var, diff, exp
>>> x, t = var("x, t")
>>> c = exp(-t)*exp(I*x)
>>> c
exp(-t)*exp(I*x)
>>> d1t = diff(c,t)
>>> d1t
-exp(-t)*exp(I*x)
>>> d1x = diff(c,x)
>>> d1x
I*exp(-t)*exp(I*x)
>>> d2x = diff(d1x, x)
>>> d2x
-exp(-t)*exp(I*x)
>>> d1t - d2x
0
```

the heat equation

A plausible model works with a decaying exponential in time:

$$c(x, t) = e^{-t} e^{lx}, \quad l = \sqrt{-1}.$$

Then we have

$$\frac{\partial c}{\partial t} = -e^{-t} e^{lx}$$

and

$$\frac{\partial^2 c}{\partial x^2} = -e^{-t} e^{lx}, \quad \text{as } l^2 = -1.$$

Thus:

$$\frac{\partial c}{\partial t} = \frac{\partial^2 c}{\partial x^2}$$

is an example of *the heat equation*.

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the Laplacian operator

Definition (the Laplacian operator)

The *Laplacian operator* ∇^2 applied to $f = f(x, y, z)$ is $\nabla \cdot \nabla f$.

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} \text{ is the gradient vector of } f.$$

$$\begin{aligned} \nabla \cdot \nabla f &= \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot \nabla f \\ &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \end{aligned}$$

The heat equation, or the equation of diffusion is then

$$\frac{\partial u}{\partial t} = \nabla^2 u, \quad u = u(x, y, z, t).$$

the Laplacian and edge detection

Consider $f(x, y)$ as the value of the gray scale in a picture.

The problem is to detect the edges.

Exercise 2:

Use divided differences for the second derivatives in

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}.$$

Why is the application of this operator useful to detect edges?

Exercise 3:

What is the Laplacian operator in polar coordinates?

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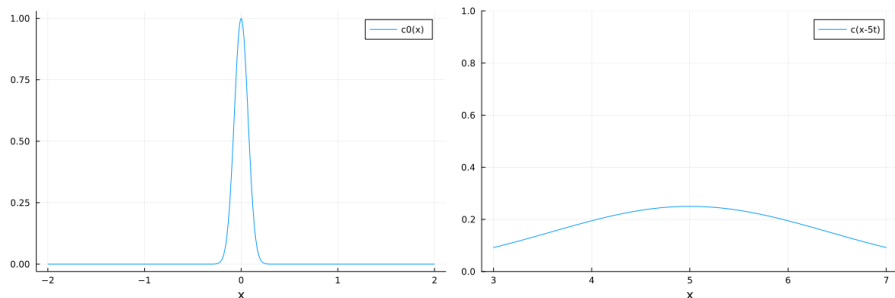
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the advection diffusion equation

Consider the evolution of the profile of fumes:



The transition from high to low concentration is modeled by

$$\frac{\partial c}{\partial t}(x, t) + \frac{\partial(v c)}{\partial x}(x, t) = k \frac{\partial^2 c}{\partial x^2}, \quad k > 0, \quad c(x, 0) = c_0(x).$$

three dimensional advection diffusion

For $c = c(x_1, x_2, x_3, t)$, the advection diffusion equation is

$$\frac{\partial c}{\partial t} + \nabla \cdot (\vec{v} c) = \sum_{i=1}^3 \frac{\partial}{\partial x_i} \sum_{j=1}^3 \left(k_{i,j} \frac{\partial c}{\partial x_j} \right),$$

where

- $k = (k_{i,j})$ is a positive definite matrix, and
- $c(x_1, x_2, x_3, 0) = c_0(x_1, x_2, x_3)$ is the initial concentration.

numerically solve the advection diffusion equation

The advection diffusion equation is solved in chapter 2 of *Industrial Mathematics: A Course in Solving Real-World Problems* by Avner Friedman and Walter Littman, published by SIAM in 1994.

Proposal for a Topic of a Project:

Study chapter 2 of the Friedman-Littman book.

Review section 8.1 of *Numerical Analysis* of T. Sauer (MCS 471).

Do the following:

- Describe the solution using your own words.
- Do the computational experiments to examine the stability.
- Write a report on your computational experiments.

Fick's laws of diffusion

The heat equation was derived by Adolf Fick in 1855.

Proposal for a Topic of a Project:

Study the wikipedia page on the derivation of Fick's second law and consult the references.

Do the following:

- Describe the derivation of the heat equation.
- Reproduce the illustration on the wikipedia page on Fick's laws of diffusion.
- Write a report on your computational experiments.

population density of age segment

We are interested to model the population of an age segment.

Proposal for a Topic of a Project:

Let $u = u(x, t)$ be the population density defined as

$$P(b, t) - P(a, t) = \int_a^b u(x, t) dx$$

the population at time t between the ages a and b . Let $q(x)$ be the probability of death less net immigration at age x per year.

- Show that $u_t + u_x + qu = 0$.
- Solve analytically (see page 227 of the text book).
- Simulate the model numerically and interpret the results.

summary and bibliography

We derived the transport equation.

Adding diffusion to it gave the advection diffusion equation.

We started chapter 11 of our text book.

- Charles R. MacCluer:

Industrial Mathematics. Modeling in Industry, Science, and Government. Prentice Hall, 2000.

The air quality modeling problem followed chapter 2 of

- Avner Friedman and Walter Littman:

Industrial Mathematics: A Course in Solving Real-World Problems, SIAM 1994.